

- Hormone receptor complex forms inside cytoplasm
- Enzymes activated inside nucleus/organelles
- Genes activated to form a particular protein
- Slower acting/effects in hours or days

Chapter 2: Nervous system

MULTIPLE-CHOICE QUESTIONS

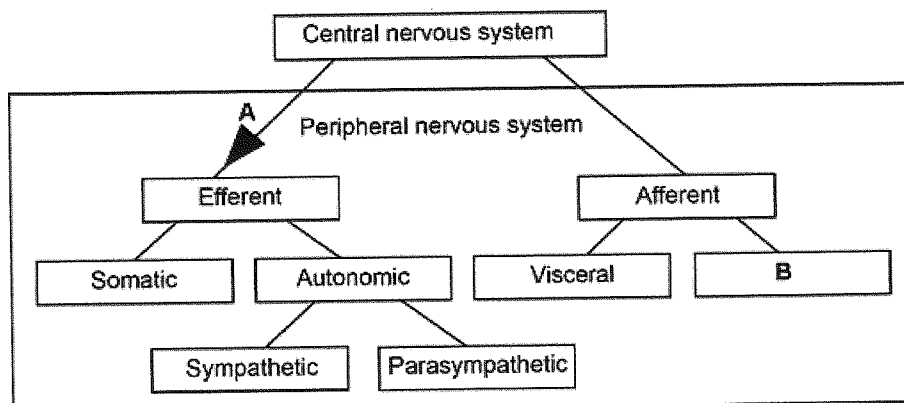
1 (2013:01)	(b)	2 (2013:14)	(c)	3 (2013:15)	(a)	4 (2013:18)	(a)
5 (2013:19)	(c)	6 (2013:29)	(d)	7 (2013:30)	(d)	8 (2014:02)	no solution available
9 (2014:03)	no solution available	10 (2014:04)	(c)	11 (2014:12)	(b)	12 (2014:18)	(b)
13 (2015:14)	(c)	14 (2015:15)	(a)	15 (2015:16)	(c)	16 (2015:25)	(c)
17 (2015:26)	(a)	18 (2016:01)	(b)	19 (2016:02)	(d)	20 (2016:10)	(a)
21 (2016:14)	(b)	22 (2016:15)	(a)	23 (2016:16)	(d)	24 (2016:26)	(a)
25 (2016:30)	(d)	26 (2017:01)	(d)	27 (2017:02)	(a)	28 (2017:13)	(a)
29 (2017:19)	(b)	30 (2017:20)	(b)	31 (2017:23)	(d)	32 (2017:30)	(d)
33 (2018:01)	(c)	34 (2018:06)	(c)	35 (2018:07)	(a)	36 (2018:08)	(b)
37 (2018:12)	(a)	38 (2018:14)	(c)	39 (2018:15)	(a)	40 (2018:17)	(d)
41 (2018:18)	(c)	42 (2018:26)	(d)				

SHORT ANSWER QUESTIONS

43 2013:39

(13 marks)

- (a) (i) Motor/efferent/myelinated/effector (1)
(ii) White matter/ascending tracts/descending tracts (1)
- (b) (i) Arrow must travel from the skin to the Muscle fibre. (1)
(ii) Any three of: (1 mark each - max. 3) (3)
• Rapid/involves only a small number of neurons
• Automatic/Involuntary/without conscious thought
• Requires a stimulus/not spontaneous
• Stereotyped/happens the same way every time
- (iii) Protective/prevents (further) damage/allows removal of danger before being aware. (1)
- (c) Skin (1)
Receptors for pain are at the ends of/in the sensory neuron (1)
- (d) (i) Arrow head going away from CNS (1)



- (ii) 1 mark both points (1)
 • From: Skin/muscles/exteroceptors/proprioceptors/bones/joints
 • To: the CNS/brain
- (iii) Each point – 1 mark each (max. 2) (2)
 • Sympathetic
 • Excessive sweating

44 2014:36

(10 marks)

- (a) A – hypothalamus (3)
 B – Cerebrum/cerebral cortex/parietal lobe/sulci
 C – cerebellum
- (b) (i) Cerebellum/C (1)
 (ii) Any two of: (2)
 • Lack of balance
 • Lack of muscle tone/reaction speed increases/floppiness
 • Reduced speed of movements
 • Inability to estimate how much time has passed
 • Inability to compare actual and intended movement
 CANNOT say anything about fine motor control/shaking/shaky movement/tremor as this is already implied in the question
- (iii) • Conscious/motor control of movement is in the cerebrum (so not affected) (1)
 • Cerebrum doesn't need to receive sensory information from outside the brain/the cerebellum needs to receive sensory information from outside the brain/cerebrum can bypass cerebellum (1) (2)
- (c) One mark for name and ONE mark for explanation: (2)
 Cranium/skull
 Hardness of cranium/bone protects/shape of skull is rounded to reflect blows/encases brain
 OR
 Meninges
 Outer layer/dura mater is tough/inflexible preventing brain movement/inner layer/pia mater seals against infection
 OR
 Cerebrospinal fluid/CSF
 Cushions brains/acts as shock absorber

45 2014:39

(6 marks)

- (a) Node of Ranvier (1)
 (b) Any ONE of the following: (1)
 • Insulate axon/fibre
 • Protect axon/fibre from damage
 • Faster nervous transmission
 • Saltatory conduction/propagation of impulse
 • Produce myelin
- (c) • (Depolarisation) occurs at Nodes of Ranvier along fibre/occurs at X/ jumps from node to node (1) (4)
 • (Sodium channels open for) sodium ions to diffuse in across membrane/into/inside axon cytoplasm
 • (following depolarisation sodium channels close and potassium channels open for) potassium ions to diffuse out across membrane/outside membrane
 • Inside membrane becomes negative in relation to outside
 • A sodium-potassium pump transports sodium ions out of the cell and potassium ions in to return to a polarised/resting state

46 2015:32

(15 marks)

- (a) One of the following: (1)
 • Mostly occurs in elderly/old age (1)
 • Caused by deficiency of neurotransmitters (dopamine in Parkinsons and ACh in Alzheimers) (1)
 • Some memory loss (only in elderly)
- (b) (i) 3 (1) (1)
 (ii) Any two of the following points (2)
 • Nerve/contains neuron fibres
 • Both sensory and motor fibres/mixed nerve/afferent and efferent fibres
 • Nerve surrounded by connective tissue/epineurium

- (iii) One mark for each part of the pathway (in order): (4)
 • Sensory/afferent neuron carries impulse (stimulus from receptor) to spinal cord
 • Passes impulse to connector/association neuron
 • Motor/efferent neuron carries impulse out of spinal cord
 • Impulse received by effector (for response) (3)
- (c) One mark for each part of the process (in order): (3)
 • Calcium released in pre-synaptic knob/axonal ending/exocytosis of neurotransmitter/s from vesicles
 • Diffusion of neurotransmitter across synapse/synaptic gap
 • Neurotransmitter binds to specific receptors on post-synaptic membrane/ on dendrite of next neuron
- (d) Must provide TWO related differences (1 mark for each statement for **autonomic** = 2 marks AND 1 mark for each statement for **somatic** = 2 marks) (4)

Autonomic	Somatic
Effectors include smooth muscles, cardiac muscle/heart, glands	Effectors are skeletal muscles
Involuntary/not under conscious control	Voluntary/conscious control
Two nerve fibres travel from CNS to effector/ preganglionic and postganglionic fibres/synapse at ganglion	Only one nerve fibre from CNS to effectors/no ganglion
Two neurotransmitters involved/noradrenaline and acetylcholine	Neurotransmitter is acetylcholine
Sympathetic and parasympathetic fibres/excitatory and inhibitory	One fibre (motor) to effector/excitatory

(11 marks)

47 2016:32

(1)

- (a) any 1 of the following (3)
 • Pain
 • Touch
 • Pressure
 • Non encapsulated
- (b) 1 mark for each dot point (1)
 • Close to surface of skin
 • Can detect small changes (pressure/temperature) quickly
 • Rapid detection helps protect body
- (c) any 1 of the following (2)
 • Rise or fall in temperature
 • Pain
 • Pressure
- (d) 1 mark for each (1)
 Afferent neuron – B
 Interneuron – A
- (e) Cell bodies of sensory/afferent neurons (3)
 (f) 1 mark for any 3 dot points
 • Stimulus is required
 • Protects from tissue damage
 • Rapid response
 • Stereotyped response
 • Involuntary response

48 2017:31

(9 marks)

- (a) Sensory/receptor (neuron) (1)
 (b) Motor neurons and interneurons have one axon and multiple dendrites extending from the cell body/cell body is between axon and dendrites (1)
 Sensory neuron has a cell body to one side of its axon/only the axon is attached to cell body, dendrites do not emerge from the cell body (1)

- (c) One mark for each point in the table (6)

	Nervous System	Endocrine System
Speed of action	Nerve impulses travel rapidly (response may occur in milliseconds)	Hormones travel at the speed of the blood (response may take several seconds or days)
Duration of action	When stimulus ceases – response ceases almost straight away	Response may last a long time, even years
Specificity of message	Nerve impulses travel along a nerve fibre to a specific part of the body – often to one effector	Hormones travel to all parts of the body, often affect a number of organs or tissues

- 49 2017:32 (6 marks)
(c) One mark for each point in the table

Zombie Symptom	Structure/s of the nervous system affected	Reason
Jerky movements when walking	Cerebellum	A healthy cerebellum coordinates muscular movements, if damaged those movements are no longer smooth
Reduced perception of pain stimulus	Cerebral cortex/ cerebrum	Several parts of the cerebral cortex process information from pain receptors
Inability to speak	Cerebral cortex/ Cerebrum	Speech is largely controlled by the left hemisphere in the temporal lobe of the cerebrum

- 50 2018:31 (12 marks)
(a) One mark for each point in the table (3)

Label	Structure name
A	dendrites
B	myelin sheath/Swann cell/myelinated axon
C	unmyelinated axon

- (b) One mark for each point (2)
 - Neuron 1 is myelinated, therefore impulses jump from one node of Ranvier to the next and do not have to pass along every section of the axon cylinder. This is called saltatory conduction.
 - Neuron 2 is not myelinated, therefore impulses moves along the entire axon cylinder. It takes longer for the depolarization to reach the axon terminals.

(c) (i)
 - Skeletal muscles/voluntary muscles (1)
(ii) One mark for each point (2)
 - Somatic pathway – one nerve fibre from CNS to the muscle.
 - Autonomic pathway – two nerve fibres from CNS to muscle or gland with a synapse in a ganglion.

(d) (i)
 - Sympathetic (1)
(ii) One mark for each point (2)
 - Adrenal medullas release increased amounts of adrenaline and noradrenaline. These two hormones target the tissues and organs that bring about the “flight or fight response”.
 - One of the responses – blood vessels that service the intestines and stomach constrict, less blood flows to these organs – this made Natasha feel nauseas or sick.

(iii) Any one of the following (1)
 - Heart rate and force of heart contractions increase – increase blood pressure.
 - Blood vessels that service skeletal muscles, heart and liver dilate. More blood flows to these organs.

- Blood vessels that service kidneys and skin constrict. Less blood flows to these organs.
- Sweat glands produce more sweat.
- Bronchioles dilate allowing more air to flow in and out of the lungs.
- Liver breaks down more glycogen increasing the blood sugar level.

- 51 2018:34 (1 mark)
(b)
 - neurotransmitters are chemicals that diffuse across a synapse that separates one neuron from another, from an axon to a dendrite, that enable a message to pass from one neuron to the next. (1)

EXTENDED ANSWER QUESTIONS

- 52 2017:42 (10 marks)

- (a)
How a nerve impulse is propagated along a myelinated nerve fibre (8)
One mark each for any 8 of the following points
 - In a resting neuron there is a net positive charge on the outside of the membrane and a negative charge on the inside (a potential difference of – 70 mV). The membrane is described as polarised
 - There is a high concentration of positive sodium ions on the outside and a high concentration of positive potassium ions on the inside. Large negatively charged organic molecules are also located on the inside
 - Sodium ions are pumped out by the membrane and potassium ions are pumped in. The sodium-potassium pump maintains the potential difference of – 70 mV when the neuron is resting
 - Protein channels in the membrane allow sodium ions out and potassium ions in
 - The channels allow potassium ions to leak out more quickly than sodium ion can leak in. This ensures a negative membrane potential in a resting neuron
 - When an impulse travels along the neuron sodium channels are opened
 - Sodium ions rush into the neuron causing a switch in the membrane potential – it becomes negative on the outside and positive on the inside (+30 mV)
 - A small depolarisation on the edge of the impulse causes more sodium channels to open in front of the impulse – this causes the spike in the impulse to move down the neuron
 - If sufficient depolarisation (15 mV) occurs – (a threshold potential) causes a wave of depolarisation – the impulse self-propagates
 - Depolarisation also causes the potassium channels to open – therefore the potassium ions leave the neuron moving rapidly out of the neuron causing the membrane to be repolarised.
 - The impulses move in one direction along the neuron because the sodium channels cannot immediately reopen. There is a refractory period during which the depolarisation cannot occur.

- Propagation of a nerve impulse in an unmyelinated nerve fibre** (2)
One mark for each of the following points
 - A myelin sheath enables the action potential to jump from one node of Ranvier to the next (Saltatory conduction)

This speeds up the impulse it may travel up to 120 m s⁻¹
 - In an unmyelinated membrane the depolarisation must occur all along the fibre, therefore it takes longer to travel along an unmyelinated fibre (max speed ~ 2 m s⁻¹) than a myelinated fibre

Chapter 3: Homeostasis

Temperature

MULTIPLE-CHOICE QUESTIONS

1 (2014:10)	(d)
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- PCR can be used to amplify a small sample of a virus.
- Early detection of a pathogenic virus enables appropriate early medical intervention to reduce the impact of the infection.

12 2018:34

(2 marks)

- (a) One mark each for any two of the following
- Biochemistry is the study of the chemistry associated with living things. Comparative biochemistry involves comparing the DNA, proteins and enzymes of organisms in order to determine their evolutionary relationships.
 - Comparing the primate neurotransmitters in primates indicates that there are close similarities between humans and other primates. This supports the comparisons in DNA, proteins and enzymes which indicate a close relationship.
 - While the levels of dopamine and acetylcholine in humans were different from the other primates, the fact that they shared these neurotransmitters supports the theory that they are closely related.

EXTENDED ANSWER QUESTIONS

13 2014:41

(12 marks)

- (b) 4 marks for each PCR step in the process:
3 marks for each step description

Step 1 - Denaturation (1)

- Heating of double stranded DNA (94–98°C)
- Disrupts hydrogen bonds between complementary bases
- Separation of the strands/single strand of DNA produced

Step 2 - Annealing/Hybridisation (1)

Any three of:

- Temperature is lowered (50–65°C)
- Primer is a small single strand of DNA
- DNA strand binds to complementary base sequences (on primer)
- DNA is annealed to a primer
- Starts the replication

Step 3 - Extension/Elongation/Synthesis (1)

Any three of:

- Taq/DNA polymerase binds to the primer (DNA template hybrid/combined primer and single DNA strand)
- Polymerase synthesizes a new DNA strand (adds nucleotides in a sequential manner)
- Complementary to the DNA template strand
- Segments of single stranded DNA are replicated
- Through a repeated series of cycles/this process is repeated many times/a series of cycles of heating and cooling.

Chapter 10: Science Inquiry Skills

MULTIPLE-CHOICE QUESTIONS

1 (2015:21)	(c)	2 (2015:22)	(d)	3 (2017:04)	(b)	4 (2017:15)	(c)
5 (2018:09)	(a)	6 (2018:29)	(c)	7 (2018:30)	(d)		

SHORT ANSWER QUESTIONS

8 2013:36

(13 marks)

(a) One mark for each correct component – max. 5

(5)

Description	Marks
Title	1
Time in first column (either as minutes or time of day)	1
Appropriate column headings with (core) body temperature as columns two and three	1
Correct units in the heading/data only in the cells – no units	1
Data in cells correctly transferred from list 'notebook'	1

Title : Core body temperature following administration of (placebo and) MDMA

Time (min)	Core body temperature with MDMA (°C)	Core body temperature with placebo (°C)
0	36.9	36.9
60	37.1	37.0
120	37.5	37.0
180	37.6	37.1
240	37.6	37.1

OR

Time of day (hours)	Core body temperature with MDMA (°C)	Core body temperature with placebo (°C)
11am/11.00	36.9	36.9
12pm/noon/12.00	37.1	37.0
1pm/13.00	37.5	37.0
2pm/14.00	37.6	37.1
3pm/15.00	37.6	37.1

- (b) MDMA/ecstasy increases (core) body temperature above normal/any statement that includes directional change between the independent variable (MDMA) and the dependent variable (core body temperature) or nil/null hypothesis (1)
- (c) Any one of: (1 mark each – max. 1) (1)
- Standardise amount of MDMA for each participant
 - Control for the effect of body mass
 - All participants receive the same relative amount of drug
 - All participants receive the same amount of drug per unit body mass
 - Dose of a drug depends on body weight
 - Make it a fair test
- (d) (i) Any two of: (1 mark each – max. 2) (2)
- Number of times MDMA and placebo administered
 - Same method of administration of MDMA and placebo (tablet)
 - Room temperature
 - Same amount of time from administration of MDMA/placebo to recording core body temperature
 - Period of time over which participants were assembled/core body temperature recordings were taken
 - All participants had both MDMA and placebo

(ii) Any one of: (1 mark each – max. 1)

(1)

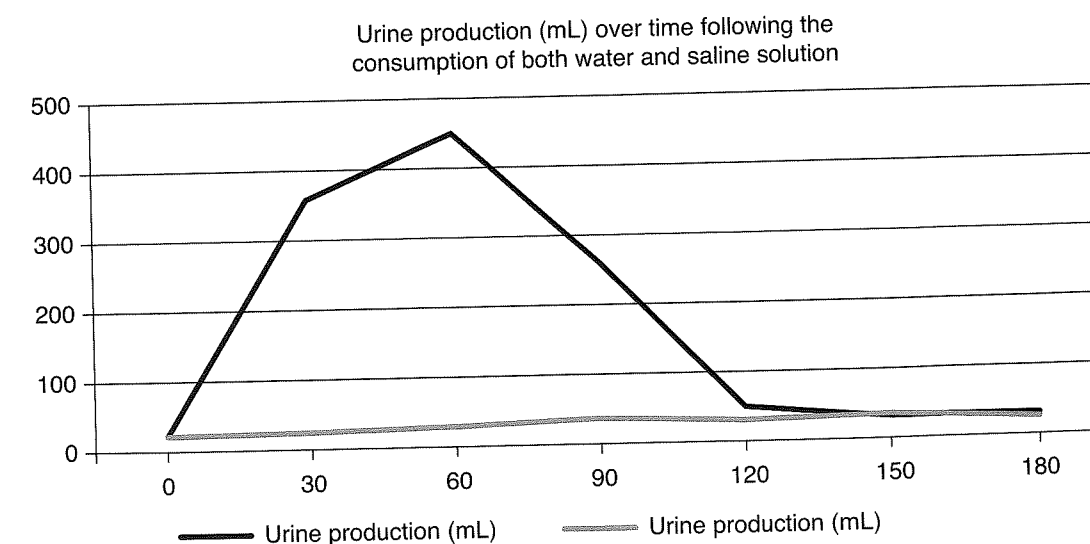
Describing the effect of the stated variable on <u>core body temperature</u> (either increase or decrease core body temperature)	
number of times MDMA and placebo administered	as increasing frequency of MDMA administration could increase core body temperature
same method of administration of MDMA and placebo (tablet)	as injecting MDMA could increase core body temperature more/faster than ingesting MDMA/action of the drug may be affected by digestion
room temperature	as a hotter room could increase core body temperature/a cooler room could decrease core body temperature
same amount of time from administration of MDMA/placebo to recording core body temperature	to allow core body temperature values at each time period to be validly compared/averaged
period of time over which participants were assembled/core body temperature recordings were taken	to allow comparison as core body temperature of participants could increase/decrease over different time periods
all participants had both MDMA and placebo	to control any effect of participants' health, diet, etc on core body temperature

- (e) Increased rate of (cellular) respiration/increased metabolic rate/metabolism (1)
- (f) (i) Any one of: (1 mark each – max. 1) (1)
- Heat gain from warm/warmer/ambient/external/room temperature
 - Heat transfer by radiation
- (ii) Delayed/decreased sweating after MDMA (1)

9 2014:35

(10 marks)

- (a) (i) Statement must include the independent and dependent variables (1 mark)
- For example – consuming saline solution will decrease urine production
OR consumption of water will result in an increase in urine production (2 marks)
- (ii) Any TWO of the following:
- volume/amount of fluid consumed
 - length of time of recording urine production
 - length of time to consume fluid
 - temperature (25°C) for duration of experiment
 - temperature for both groups
 - amount of/minimal physical activity for both groups
- (b) (i) Establish more controlled variables/improve experimental design/specific answer referring to the experiment (MUST improve the design) (1 mark)
- (ii) Repeat experiment again/replicate/increase sample size (1 mark)
- (c) Graph: 1 mark each for (5 marks)



Creelman Exam Questions: Human Biology ATAR Course Units 3 and 4, 2019

- title, must include the independent and dependent variables
- labelling of axes with correct units independent variable (time, minutes) on X axis/
dependent variable (urine production, mL) on Y axis
- use a suitable scale
- key/legend for lines
- correctly/accurately plots points and joins points (with ruler) to form a line

10 2015:36

(9 marks)

- (a) Any two of the following: (2)
- Delivery/administration/oral administration of both drug and placebo
 - Dose of drug and placebo given/delivered daily
 - Age range of infants/all infants were born between 26 and 28 weeks gestation
 - Length of time for treatment/treatment lasted 6 weeks
- (b) (i) Substance without active ingredient/ingredient/that has no effect/has no physiological/
pharmacological effect (1)
- (ii) Any one of: (1)
- Acts as a control/for comparison with the experimental group
 - Reduces the effects of psychological influences/effects on the results/between the two groups
- (c) Any one of the following: (1)
- Caffeine reduced the need for a breathing tube (intubation) by (approximately) one week
 - Caffeine reduced the need for additional oxygen by approximately 1.5/1.6 weeks
- (Use the last two rows of the table to ascertain this information)
- (d) Statement must include the independent and dependent variables/nil hypothesis/null hypothesis (1)
- For example – caffeine slows weight gain in premature infants. Note that weight must be the dependent variable.
- (e) (i) Carbon dioxide/ CO_2 (1)
- (ii) Cerebral cortex/cerebrum (voluntary control!) (1)
- (iii) Protection/prevents water entering lungs when swimming/in water/prevents dangerous/
poisonous/unwanted gases entering lungs (1)

11 2016:33

(5 marks)

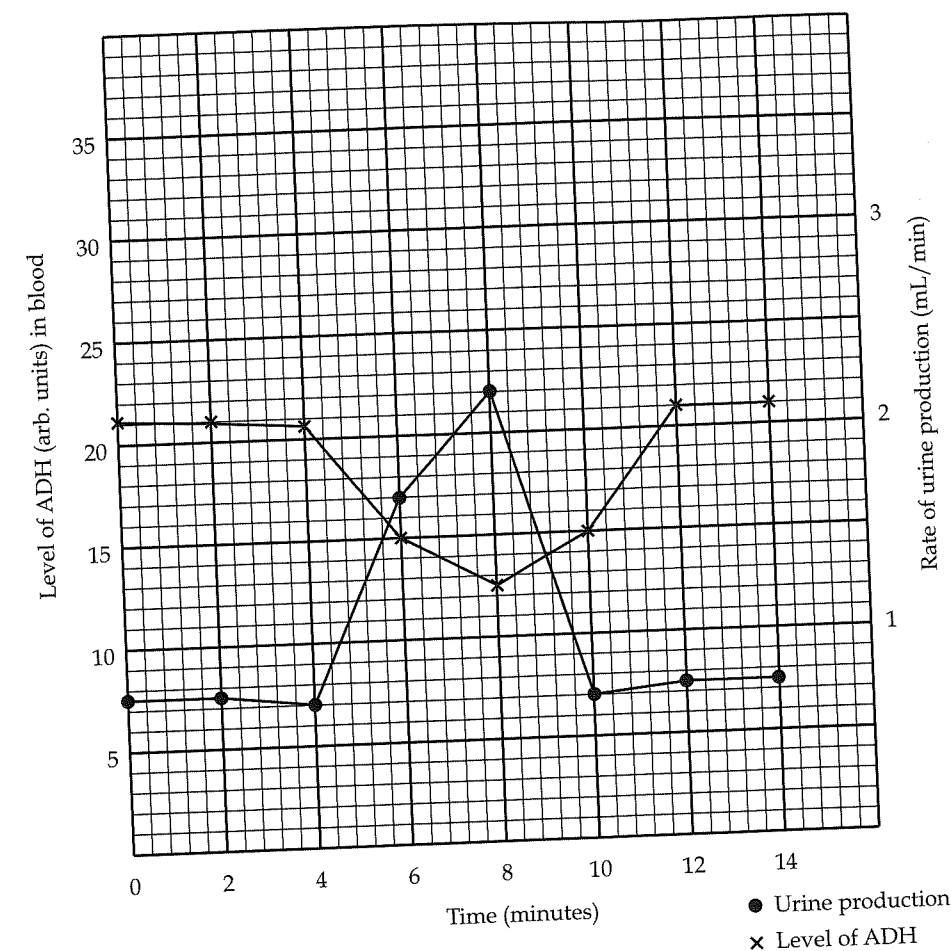
- (c) 1 mark for each dot point (2)
- Same number of diabetic rats as in the experimental group
 - Control group injected with placebo/isotonic solution/non insulin coding DNA
- (d) any 1 of the following (1)
- Daily fluctuations in blood glucose levels in normal healthy rats
 - Daily fluctuations in blood glucose levels in diabetic rats before the therapy
- (f) 1 mark for any 2 dot points (2)
- Determine side effects of treatment in rats/effect on mouse longevity
 - Repeat procedures to ensure results are consistent
 - Carry out clinical trials on humans
 - Obtain approval from appropriate health authorities

12 2016:36

(6 marks)

- (a) Graph; mark allocation 1 mark for each of the following dot points (5)
- Title – including both independent and dependent variables
 - Correctly labelled x axis (time with unit) and axis divided evenly
 - Correctly labelled y axis (dependent variable with unit) and axis divided evenly
 - Accurately positioned points and adjacent points joined by straight lines
 - Large use of over half of the grid provided and use of a key to identify clearly each line

Level of ADH in blood and rate of urine production vs time



- (b) Any answer in range 19–20 arbitrary units (1)

13 2016:39

(6 marks)

- (a) 1 mark for each dot point (3)
- Denaturing – DNA strands are separated along their lengths using heat (about 90°C)
 - Annealing – As the denatured DNA is cooled (between $48\text{--}65^\circ\text{C}$) primers, short fragments of DNA, attach to them allowing DNA polymerase to begin the next stage
 - Extension (elongation) – DNA polymerase adds free nucleotides to the remaining bases of the DNA strands (between 68 and 72°C)
- (b) 1 mark for either dot point (1)
- It creates a greater amount (amplifies) of DNA for analysis
 - Often fossil DNA is found in tiny quantities
- (c) (i) The banding of the DNA fragments provides a DNA profile of the hominid fossil (1)
- (ii) *Homo neanderthalensis* (1)

(11 marks)

14 2017:33

- (a) One mark each for any three points in the table (3)

• The vertical/y axis not labelled
• Title does not describe both the independent and dependent variables
• The lines are not labelled/no key
• Points are not accurately placed at time 0 minutes

(2)

- (b) One mark for each of the two points
- So that each group was standardised and
 - The effect of the different solutions could be compared

- (c) One mark for any two points (2)
- Average health of each volunteer needs to be similar (no diabetics)
 - Activity level similar – no difference in sweating
 - Diets similar – no other extra source of fluid
- (d) The osmotic pressure of the blood is increased/Salt concentration of the blood is increased (1)
- (e) One mark for each of the three points (3)
- 30–60 minutes after drinking the distilled water
 - At 60 mins the urine production is at its highest, this indicates low levels of ADH (which causes water retention or reabsorption)
 - As hormones generally take some time to act on their target (in this case the distal convoluted tubules and collecting ducts in the nephrons), the ADH level would be lowest before 60 minutes

15 2017:35

(5 marks)

- (a) One mark for each point

What is required for Step X to proceed?	• The DNA needs to be denatured – heating to 94–95°C
What happens to the DNA at Step X?	• The DNA is denatured – hydrogen bonds are broken producing two single strands of DNA
What is the process occurring at Step Y?	• Primers attach to each single strand – annealing at 50–56°C
What is added at Step Z?	• Nucleotides attach to their complementary bases
At what temperature does step Z occur?	• 72°C (extending stage)

16 2018:33

(9 marks)

- (a) One mark for each point (2)
- Title – indicating clearly which variables are being plotted.
 - X axis should be labelled clearly with 'time since PCR started' or similar label.
- (b) (i) • Elongation / extension. (1)
- (ii) One mark for each point (2)
- Mixture is cooled to about 55°C
 - Annealing of primers to each template strand. / Primers attach to both template strands produced in step 1 (the denaturing stage A).
- (d) One mark for each point (4)
- DNA samples are placed in wells on one side of a gel. An electric charge is turned on that pulls the fragments through the gel.
 - Because the DNA molecules are negatively charged, they move towards the positive electrode.
 - Smaller fragments move faster through the gel than larger ones, the fragments form bands across the gel.
 - When the DNA-fragments are stained, they can be seen as bands. Each band represents a group of same-sized DNA fragments.

17 2018:38

(7 marks)

- (a) • The use of the bronchodilator causes an increase in blood oxygen concentration in patients suffering from chronic asthma. (1)
- (b) One mark for each point (2)
- Independent variable – the use of the bronchodilator.
 - Dependent variable – blood oxygen concentration.
- (c) • To provide a control / to allow comparisons of blood oxygen concentration with normal conditions when no bronchodilator is used / to enable the effect of the bronchodilator on blood oxygen concentration to be determined. (1)
- (d) Any three points for one mark each (3)
- All patients should be selected at random.
 - Both groups of similar age.
 - Both groups of similar gender mix.
 - Both groups, apart from having chronic asthma, should be of similar health and fitness.
 - Both groups have a similar average size and stature.
 - Both groups have similar activities and life style during the experiment.
 - Both groups be of similar ethnicity.

Acknowledgements

Chapter	Question Reference and Acknowledgement
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